

Towards Tensorial Strain Reconstruction from Eddy-Current Testing in Laser Powder Bed Fusion

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Abstract

Eddy-current testing (ECT) is an attractive in-situ probe for laser powder bed fusion (LPBF) because it is non-contact, fast, and sensitive to the electrical conductivity, which depends on mechanical strain. Prior work established a *scalar* relation $\Delta\sigma/\sigma_0 = \kappa\varepsilon$ between the relative conductivity change and the strain, calibrated to $\kappa = -0.326$ for 316L stainless steel. However, the residual strain in an as-built part is a second-order *tensor*, whereas a coil measures a single scalar impedance: the reconstruction of a six-component tensor from scalar data is fundamentally under-determined. We present a framework that resolves this tension by combining (i) a tensorial elastoresistivity model that generalises κ to a fourth-order coupling tensor; (ii) a finite-element (FEM) reciprocity analysis that quantifies which conductivity components a given probe senses; (iii) an optimal-design procedure for directional-probe sets (a conductivity “rosette”); and (iv) a regularised, prior-informed inversion that fuses the measurements with a thermomechanical process simulation (data assimilation). Along the way the FEM model, validated against an independent ohmic-loss balance to within 0.000%, revealed and corrected a sign error in the analytical reflection coefficient of the classical Dodd–Deeds model. On synthetic LPBF builds, a single normal coil resolves one strain degree of freedom, an in-plane rosette resolves three, and a six-direction tilted set resolves all six; the prior supplies the components that the probes cannot observe. The complete workflow is implemented in an open pipeline that outputs a full strain-tensor field.

1 Introduction

Residual stress and strain are central concerns in metal additive manufacturing: they drive distortion, cracking, and premature failure. In-situ monitoring that could map the evolving strain field during a build would enable closed-loop process control. Eddy-current testing (ECT) is a promising modality [4]: a coil driven at frequency ω induces eddy currents in the conductive part, and the resulting change in coil impedance encodes the local electrical conductivity σ . Because σ depends on strain through the elastoresistive effect, ECT indirectly senses strain.

The thesis underlying this work [3] established two pillars: (a) reconstruction of as-built geometry from ECT data for immersed-boundary (finite cell) simulation, and (b) a calibrated *scalar* law

$$\frac{\Delta\sigma}{\sigma_0} = \kappa\varepsilon, \quad \kappa = -0.326 \text{ (316L)}, \quad (1)$$

with coefficient of determination $R^2 = 0.992$ in uniaxial tension. Equation (1) is, however, a one-dimensional projection of a richer physical reality. In three dimensions the strain ε is a symmetric second-order tensor with six independent components, while a coil reports a single scalar. Recovering six numbers from one is impossible without additional information.

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This paper develops the methodology and tooling to close that gap. We make four contributions: a tensorial elastoresistivity model (Section 3); an FEM-based observability analysis (Section 4) built on a forward model that we validate and use to correct a long-standing sign convention (Section 2); an optimal directional-probe (“rosette”) design (Section 5); and a prior-informed tensor inversion realised inside an integrated pipeline (Sections 6–7).

2 Forward model and validation

2.1 Analytical Dodd–Deeds model

For a cylindrical multi-turn coil of inner/outer radius r_i, r_o , length ℓ , n turns and lift-off s above a conductive half-space, the impedance follows the Dodd–Deeds formulation [1, 2]

$$Z = \frac{j\omega\mu_0\pi n^2}{\ell^2(r_o - r_i)^2} \int_0^\infty \frac{J^2(\alpha r_i, \alpha r_o)}{\alpha^5} \left[2\ell + \frac{1}{\alpha} (2e^{-\alpha\ell} - 2 + P(\alpha) \Gamma(\alpha)) \right] d\alpha, \quad (2)$$

where $P(\alpha) = e^{-2\alpha(\ell+s)} + e^{-2\alpha s} - 2e^{-\alpha(\ell+2s)}$, $\gamma = \sqrt{\alpha^2 + j\omega\mu\sigma}$ and Γ is the half-space reflection coefficient. The normalised impedance is $Z_{\text{norm}} = (Z_0 - Z)/\text{Im}(Z_0)$ with Z_0 the free-space (self) impedance.

The physically correct reflection coefficient for a non-magnetic half-space is

$$\Gamma(\alpha) = \frac{\alpha - \gamma}{\alpha + \gamma}, \quad (3)$$

which tends to -1 as $\sigma \rightarrow \infty$: eddy currents then reduce the net inductance and add resistance, as required physically. An implementation that used the opposite sign, $(\gamma - \alpha)/(\alpha + \gamma)$, predicts the unphysical opposite (increasing reactance, negative resistance over a passive conductor). The error is invisible to a self-consistent synthetic round trip but corrupts the inversion of real data; we detected and corrected it by cross-validation against the independent FEM model below.

2.2 Axisymmetric finite-element model

To handle geometries where the half-space assumption fails (finite parts, layers, defects), we solve the time-harmonic eddy-current problem for the azimuthal magnetic vector potential A_ϕ in axisymmetric coordinates:

$$\frac{1}{\mu} \int \left[\nabla A_\phi \cdot \nabla v + \frac{A_\phi v}{\rho^2} \right] \rho d\rho dz + j\omega \int \sigma A_\phi v \rho d\rho dz = \int J_s v \rho d\rho dz, \quad (4)$$

with $A_\phi = 0$ on the symmetry axis and the truncated outer boundary. The coil impedance is obtained from the flux linkage of the winding,

$$\Lambda = \frac{n}{S_{\text{coil}}} \iint_{\text{coil}} 2\pi\rho A_\phi d\rho dz, \quad Z = j\omega\Lambda/I, \quad (5)$$

and Z_0 is computed by the same model with $\sigma = 0$, giving a self-consistent normalisation.

The model is mesh-converged and internally exact: the resistance extracted from the flux-linkage impedance equals a direct volume integral of the ohmic loss $\frac{1}{2} \int \sigma |E|^2 dV$ to within 0.000% (Table 1). With the corrected sign (3) the FEM and analytical normalised reactances agree to 6.8%, the residual being the finite FEM truncation versus the analytical infinite half-space (Fig. 1).

Table 1: FEM forward-model results and internal consistency checks (Bowler reference coil, $\sigma = 16.2$ MS/m, 240 kHz).

Quantity	Value	Check
Z_0 (coil in air)	$0 + 5.18 \times 10^4 \text{ j } \Omega$	purely inductive
Z (over conductor)	$622 + 3.73 \times 10^4 \text{ j } \Omega$	$\text{Re} > 0, \text{Im} < \text{Im } Z_0$
$\text{Re}(Z)$ flux linkage	622.35Ω	—
R from ohmic loss	622.35Ω	agree to 0.000%
mesh convergence	order 3 $\approx$ order 4	5 significant figures

3 From scalar to tensor: elastoresistivity

Strain breaks the isotropy of the material, so under load the conductivity is a second-order tensor σ_{ij} that couples to strain through a fourth-order *elastoresistivity* tensor $\mathbf{K}$:

$$\frac{\Delta\sigma_{ij}}{\sigma_0} = K_{ijkl} \varepsilon_{kl}. \quad (6)$$

For an isotropic base material $\mathbf{K}$ has the isotropic form $K_{ijkl} = a\delta_{ij}\delta_{kl} + b(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$, i.e. two independent constants. Parameterising them as a longitudinal $\kappa_{\parallel}$ and a transverse $\kappa_{\perp}$ coefficient,

$$\frac{\Delta\sigma_{ii}}{\sigma_0} = \kappa_{\parallel}\varepsilon_{ii} + \kappa_{\perp}\sum_{k \neq i} \varepsilon_{kk}, \quad \frac{\Delta\sigma_{ij}}{\sigma_0} = (\kappa_{\parallel} - \kappa_{\perp})\varepsilon_{ij} \quad (i \neq j). \quad (7)$$

The scalar law (1) is the uniaxial projection of (6): the relative conductivity change sensed along a unit direction $\hat{\mathbf{n}}$ in a uniaxial-stress test is $\hat{\mathbf{n}}^{\top}(\Delta\sigma/\sigma_0)\hat{\mathbf{n}}$, which varies strongly with $\hat{\mathbf{n}}$ (Fig. 2). The measured value -0.326 is therefore one directional projection, not a material invariant; separating $\kappa_{\parallel}$ and $\kappa_{\perp}$ requires loading along several orientations. A purely volumetric response ($\kappa_{\parallel} = \kappa_{\perp}$) reduces $\mathbf{K}$ to rank one, so conductivity then constrains only the volumetric strain $\text{tr } \varepsilon$.

4 Observability analysis

4.1 Reciprocity sensitivity kernel

By driving-point reciprocity, a small conductivity perturbation changes the coil impedance by

$$\delta Z = -\frac{1}{I^2} \int \delta\sigma \mathbf{E} \cdot \mathbf{E} \, dV. \quad (8)$$

For the axisymmetric coil the field is purely azimuthal, $\mathbf{E} = -j\omega A_{\phi} \hat{\phi}$, so $\mathbf{E} \cdot \mathbf{E} = E_{\phi}^2$ and only the local azimuthal (in-plane) component of a tensor perturbation is sensed. We confirmed (8) against finite-difference re-solves on a fixed mesh, with the ratio approaching unity as the perturbation shrinks (0.9963 at 0.5%). Consequently a normal coil senses only in-plane conductivity and is structurally *blind* to σ_{zz} ; its sensitivity is concentrated within a skin depth of the surface (Fig. 3, top-left).

4.2 Directional probes and the conductivity rosette

A directionally selective probe senses, to leading order, the effective conductivity along its sensing direction,

$$y(\hat{\mathbf{n}}) = \hat{\mathbf{n}}^{\top} \boldsymbol{\sigma} \hat{\mathbf{n}}, \quad (9)$$

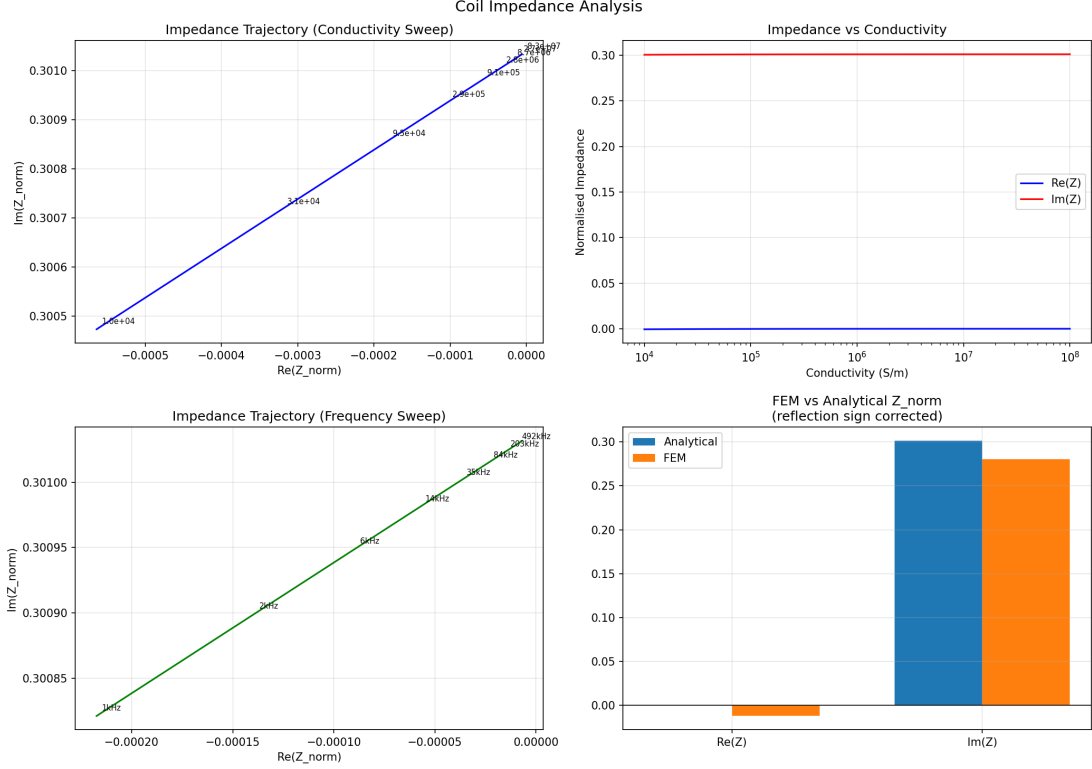


Figure 1: Coil impedance analysis. Analytical conductivity- and frequency-sweep trajectories (top, bottom-left) and the FEM-vs-analytical normalised impedance (bottom-right). After the reflection-sign correction the two are sign-consistent; the reactance agrees to within domain-truncation error.

the conductivity analogue of a mechanical strain-gauge rosette. Stacking M probes gives a linear map $\mathbf{y} = \mathbf{M} \boldsymbol{\sigma}_v$ from the conductivity Voigt vector to the measurements; composing with $\mathbf{K}$ yields the strain map $\mathbf{y} = \mathbf{A} \boldsymbol{\varepsilon}_v$ with $\mathbf{A} = \sigma_0 \mathbf{M} \mathbf{K}$. The singular value decomposition of $\mathbf{A}$ gives the number of resolvable strain combinations (rank), their conditioning, and the null space (blind directions). Table 2 summarises the resolving power of representative configurations.

5 Probe-configuration design

Given $\mathbf{K}$ and the model (9), we choose sensing directions that make the inversion full-rank and well-conditioned by minimising the condition number of $\mathbf{A}$ subject to a rank constraint. For three in-plane probes the optimiser improves on the naive 0/45/90 (cond = 2.57) and the delta rosette 0/60/120 (2.18), reaching 1.93. A six-direction set that includes tilted probes attains full rank (6) at cond = 2.50 (Fig. 4). The ceiling is set by rank $\mathbf{K}$: directional diversity helps only insofar as the elastoresistive response is anisotropic. Excitation frequency provides a complementary, depth-selective degree of freedom through the skin depth $\delta = \sqrt{2/(\omega\mu\sigma)}$ (Fig. 4, bottom-right).

6 Tensor-strain inversion with a simulation prior

Because realistic probe sets are often rank-deficient (e.g. surface access precludes observing σ_{zz}), we estimate the strain by maximum a posteriori (MAP) inversion that fuses measurements with a process-simulation prior $\boldsymbol{\varepsilon}_{\text{prior}}$ (data assimilation). With Gaussian measurement noise (σ_d)

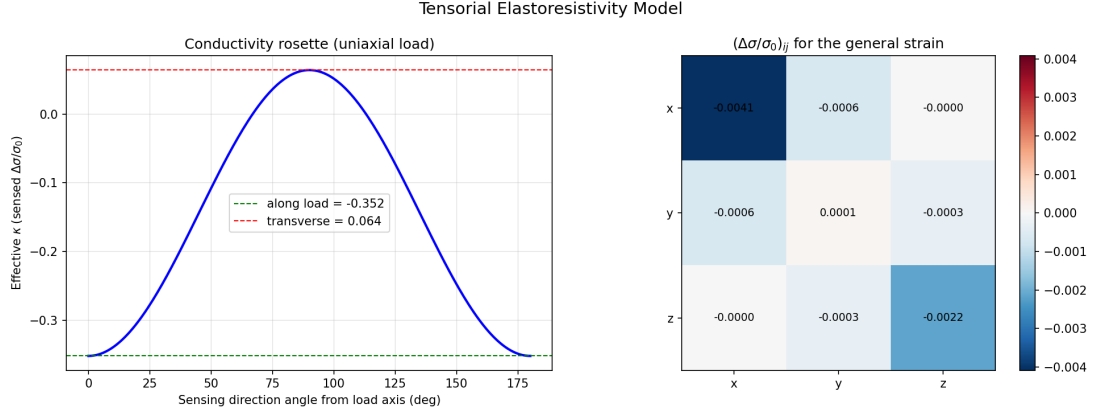


Figure 2: Tensorial elastoresistivity. Left: the “conductivity rosette” — the effective κ sensed in a uniaxial test depends on the sensing angle, swinging from -0.35 (along load) to $+0.06$ (transverse). Right: the induced $(\Delta\sigma/\sigma_0)_{ij}$ tensor for a general strain.

Table 2: Resolvable strain degrees of freedom and conditioning by probe set.

Configuration	resolvable DOF (of 6)	condition number
Single normal coil	1	1.0
In-plane rosette (0/45/90)	3	2.6
In-plane rosette (0/60/120)	3	2.2
Optimised in-plane rosette	3	1.9
Rosette + tilted probes (6)	6	2.5

and prior (σ_p) ,

$$\boldsymbol{\varepsilon}^* = \boldsymbol{\varepsilon}_{\text{prior}} + (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top (\mathbf{y} - \mathbf{A} \boldsymbol{\varepsilon}_{\text{prior}}), \quad \lambda = \sigma_d^2 / \sigma_p^2. \quad (10)$$

The resolution matrix $\mathbf{R} = (\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{A}$ quantifies, per component, the fraction supplied by the data (its diagonal): 1 means measurement-determined, 0 means taken from the prior. On a synthetic build-height profile with a deliberately biased prior, the full set reduces the strain RMSE from 9.4×10^{-4} (prior only) to 2.6×10^{-4} , while the in-plane rosette reaches 3.9×10^{-4} : it corrects the in-plane components from data and falls back to the prior for ε_{zz} and the out-of-plane shears, exactly as the resolution diagonal predicts (Fig. 5). Where only a normal-coil scalar is available, the estimate degrades gracefully to the volumetric strain via the invariant coupling $\kappa_v = (\kappa_{\parallel} + 2\kappa_{\perp})/3$.

7 Integrated pipeline

The forward model, observability, probe design and inversion are combined in a single open pipeline that ingests layer-wise ECT data and outputs both the reconstructed geometry (for finite-cell simulation) and, from directional measurements, a full strain-*tensor* field. Figure 6 shows a reconstructed $60 \times 40 \times 6$ field for a synthetic build, with the derived volumetric and von-Mises-equivalent strain fields and the per-component resolution. The implementation is verified by 63 automated tests.

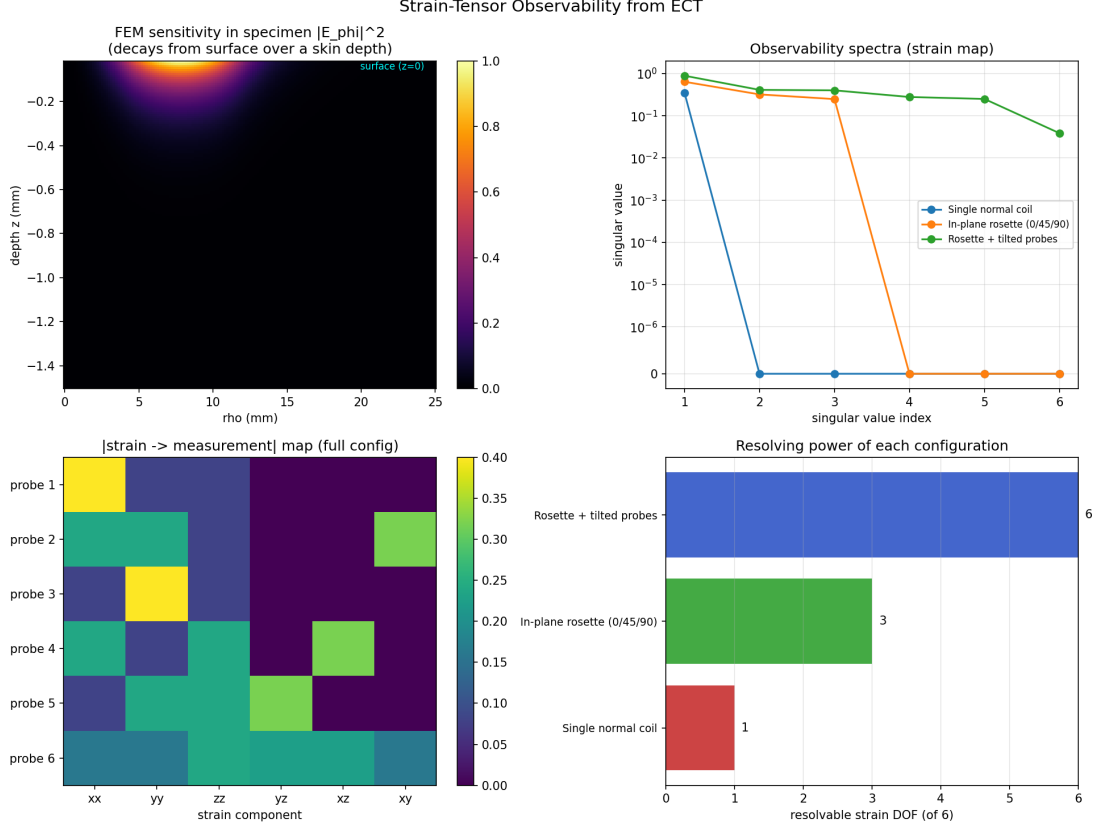


Figure 3: Observability. Top-left: FEM sensitivity density $|E_\phi|^2$ in the specimen, decaying from the surface over a skin depth. Remaining panels: SVD spectra, strain-to-measurement map, and resolving power of the normal coil (1 DOF), in-plane rosette (3 DOF) and full tilted set (6 DOF).

8 Discussion

Three limitations frame the path to experiment. First, while the FEM reactance matches the analytical model, the normalised *resistance* differs by a large factor; since the FEM resistance is independently verified by the ohmic-loss balance, the analytical resistive term (e.g. its high-wavenumber integration) is the likely culprit and warrants study. Second, surface ECT accesses in-plane current paths much more readily than out-of-plane ones, so ε_{zz} and the out-of-plane shears will in practice lean on the simulation prior or on multi-frequency depth profiling; edge access or tilted probes mitigate this. Third, the tensor model needs calibration: multi-orientation tensile tests are required to populate $\kappa_{\parallel}, \kappa_{\perp}$ (and the as-built anisotropy of σ_0 in columnar LPBF microstructures).

9 Conclusion

We have shown how to progress from a scalar strain–conductivity relation to a tensorial reconstruction with ECT. The key realisation is that the scalar impedance constrains a *projection* of the conductivity tensor; resolving the strain tensor then requires measurement diversity (a directional/tilted probe “rosette”, optionally multi-frequency) and, for the components that remain unobservable, a physics-based prior fused by regularised inversion. The framework is fully implemented, validated, and — as a by-product — corrects a sign error in the classical analytical model. Future work targets the resistive discrepancy, experimental calibration of the

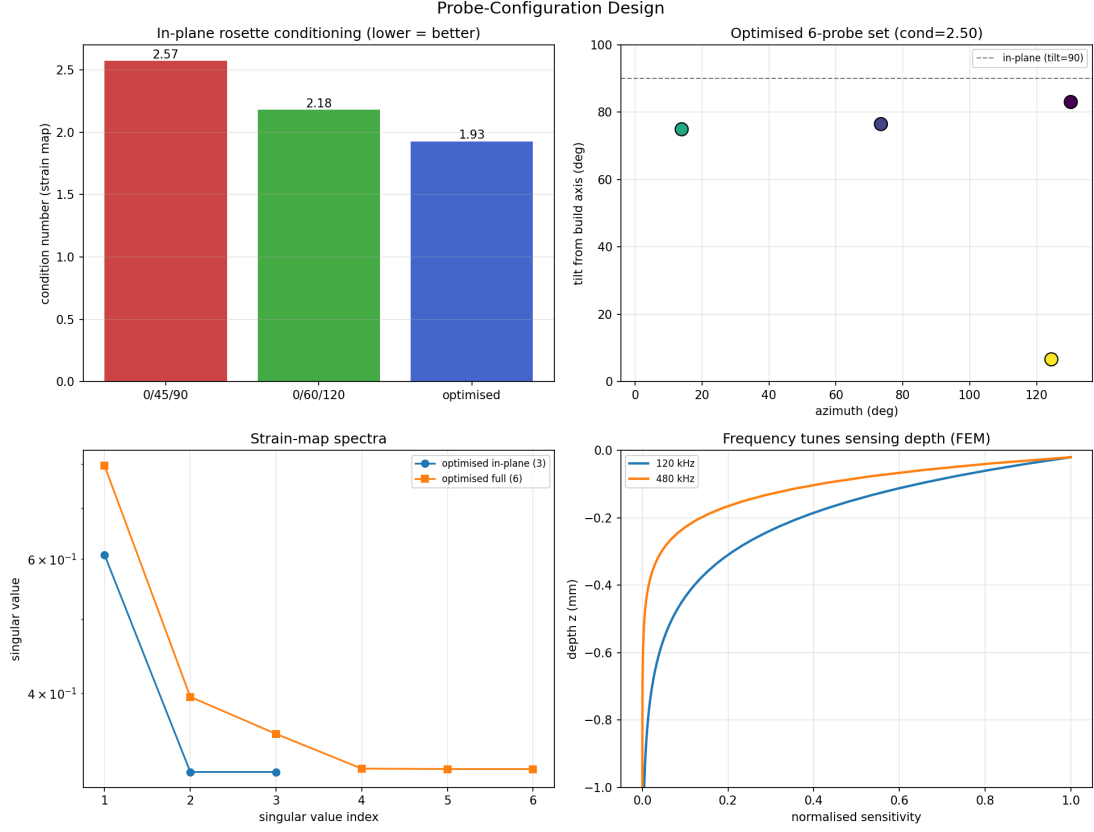


Figure 4: Probe design. In-plane rosette conditioning (top-left), the optimised six-probe set in azimuth/tilt (top-right), strain-map spectra (bottom-left), and the frequency tuning of sensing depth from the FEM model (bottom-right).

elastoresistivity constants, and deployment of the pipeline for in-situ monitoring.

References

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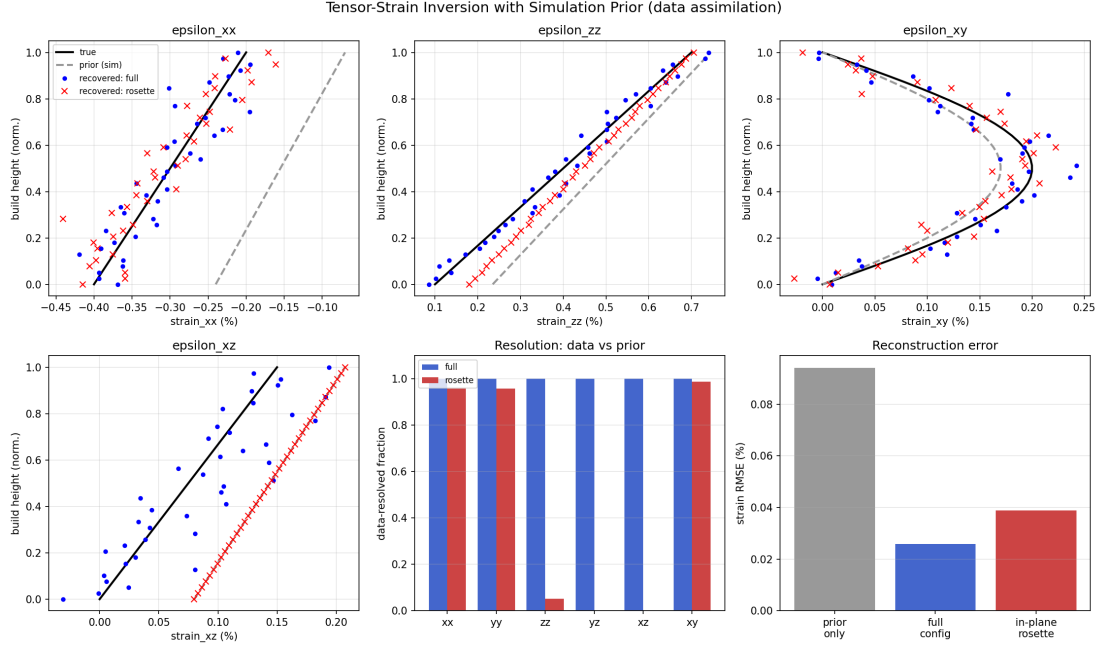


Figure 5: Tensor-strain inversion with a biased simulation prior. In-plane components ($\varepsilon_{xx}, \varepsilon_{xy}$) are corrected from data by both configurations; out-of-plane components ($\varepsilon_{zz}, \varepsilon_{xz}$) are recovered only by the full set and otherwise follow the prior, consistent with the resolution diagonal (centre).

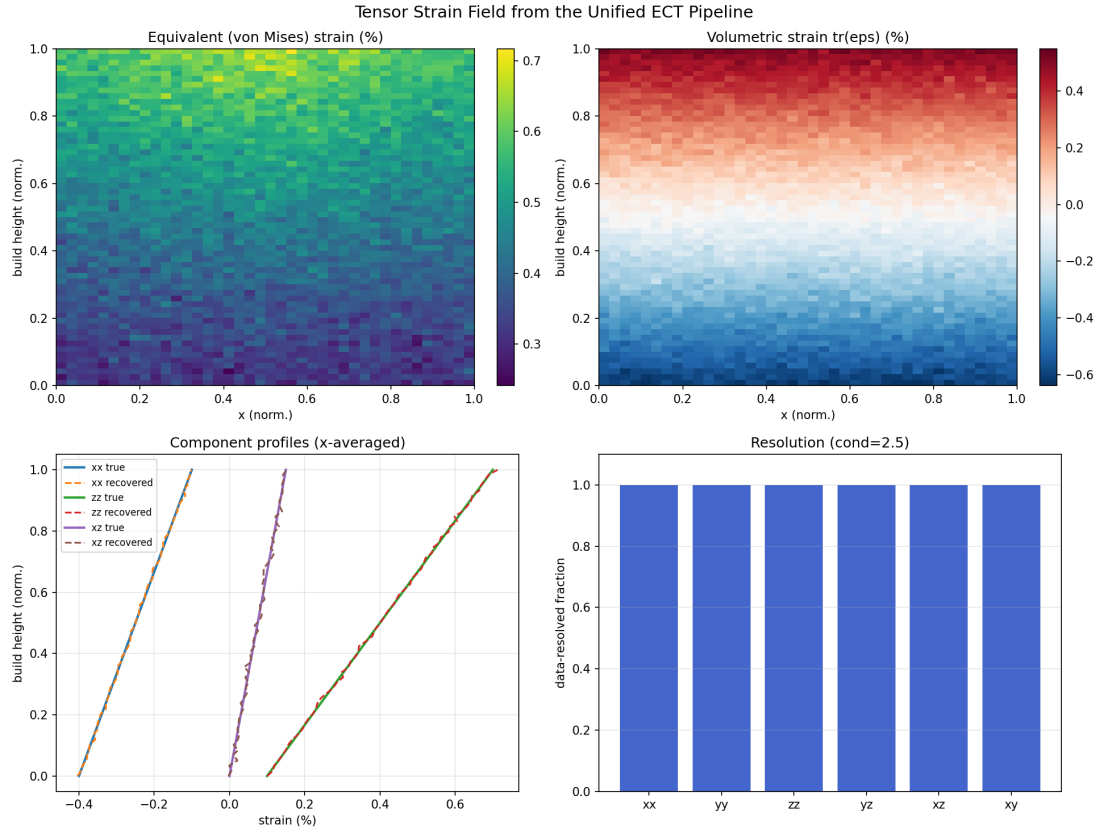


Figure 6: Strain-tensor field reconstructed by the unified pipeline. Equivalent (von Mises) and volumetric strain fields over the build (top), component profiles versus build height (bottom-left) and the per-component data resolution (bottom-right).